

Note on cocaine toxicity:

During the initial implementation of the intranasal cocaine self-administration paradigm, we observed that signs of cocaine toxicity (i.e., overdose) in some mice did not always appear acutely within a single session but could emerge after repeated. We observed that mice could tolerate relatively high cocaine intake for two consecutive sessions without any visible adverse effects. However, on the third day, prior to beginning that day's session, some animals exhibited clear signs of cocaine toxicity, including lethargy, poor coat condition, and reduced responsiveness. This pattern suggests that toxicity can emerge abruptly after repeated exposure, even when animals initially appear unaffected. To mitigate this risk, we recommend implementing conservative intake caps (e.g., 40 mg/kg/hour) during early sessions (see rolling cap Med PC code).

When implementing the intranasal cocaine self-administration paradigm, it is critical to limit the total cocaine intake during the initial training sessions to prevent overdose and adverse effects. Early sessions should employ a conservative intake cap (e.g., 40 mg/kg) to allow animals to acclimate to the procedure and to assess individual response sensitivity. Close monitoring of animal behavior is essential, and sessions should be terminated immediately if any signs of distress are observed. This cautious approach ensures animal safety while maintaining the integrity of subsequent behavioral assessments.

Acute cocaine overdose symptoms:

Tremors

Convulsions/seizures

Stereotypies (repetitive sniffing, licking, or gnawing that becomes excessive)

Labored or irregular breathing

Sudden collapse